EXPERIMENT : 11

AIM:

Experiment to understand pandas library use cases in Data science.

DESCRIPTION:

Understand data frame use cases using pandas library.

CODE:

```
import numpy as np
import pandas as pd

# Create a list
list1 = [[1, 'Ram', 60000],
         [2, 'Ravi', 30000]]

# Create DataFrame
df = pd.DataFrame(list1, columns=['Empd', 'Name', 'Salary'])

# Display DataFrame
print(df)

# Display DataFrame information
df.info()

# Read CSV file
df = pd.read_csv('/content/startup.csv')

# Display information about CSV
df.info()

# Display first 5 rows
print(df.head())

# Display last 5 rows
print(df.tail())
```

OUTPUT:

```
   Empd  Name  Salary
0     1   Ram   60000
1     2  Ravi   30000
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      object
 2   Salary  2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   RND     50 non-null     float64
 1   ADMIN   50 non-null     float64
 2   MKT     50 non-null     float64
 3   STATE   50 non-null     object
 4   PROFIT  50 non-null     float64
dtypes: float64(4), object(1)
memory usage: 2.1+ KB
         RND      ADMIN        MKT       STATE     PROFIT
0  165349.20  136897.80  471784.10    New York  192261.83
1  162597.70  151377.59  443898.53  California  191792.06
2  153441.51  101145.55  407934.54     Florida  191050.39
3  144372.41  118671.85  383199.62    New York  182901.99
4  142107.34   91391.77  366168.42     Florida  166187.94
        RND      ADMIN        MKT       STATE    PROFIT
45  1000.23  124153.04    1903.93    New York  64926.08
46  1315.46  115816.21  297114.46     Florida  49490.75
47     0.00  135426.92       0.00  California  42559.73
48   542.05   51743.15       0.00    New York  35673.41
49     0.00  116983.80   45173.06  California  14681.40
```

CODE:

```
import numpy as np
import pandas as pd

# Load the employee dataset
df = pd.read_csv('/content/Employee.csv')

# Display first and last 5 rows
print(df.head())
print(df.tail())
```

```python
# Display dataset information
df.info()

# Access the MonthlyIncome column
print(df['MonthlyIncome'])

# Check the data type
print(type(df['MonthlyIncome']))

# Calculate mean
print("Mean:", df['MonthlyIncome'].mean())

# Calculate mode
print("Mode:")
print(df['MonthlyIncome'].mode())

# Calculate variance
print("Variance:", df['MonthlyIncome'].var())

# Calculate standard deviation
print("Standard Deviation:", df['MonthlyIncome'].std())

# Statistical summary
print(df.describe())

# Summary of all columns
print(df.describe(include='all'))

# Get column names
empCol = df.columns
print(empCol)

# Convert DataFrame to NumPy array
emparray = df.values
print(emparray)

# Convert array back to DataFrame
employee_DF = pd.DataFrame(emparray, columns=empCol)
print(employee_DF)
```

OUTPUT:

```
  Age Attrition     BusinessTravel  DailyRate              Department  \
0   41       Yes      Travel_Rarely       1102                   Sales
1   49        No  Travel_Frequently        279  Research & Development
2   37       Yes      Travel_Rarely       1373  Research & Development
3   33        No  Travel_Frequently       1392  Research & Development
4   27        No      Travel_Rarely        591  Research & Development

   DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber  \
0                 1          2  Life Sciences              1               1
1                 8          1  Life Sciences              1               2
2                 2          2          Other              1               4
3                 3          4  Life Sciences              1               5
4                 2          1        Medical              1               7

   ...  RelationshipSatisfaction StandardHours  StockOptionLevel  \
0  ...                         1           80                 0
1  ...                         4           80                 1
2  ...                         2           80                 0
3  ...                         3           80                 0
4  ...                         4           80                 1

   TotalWorkingYears  TrainingTimesLastYear WorkLifeBalance  YearsAtCompany  \
0                  8                      0               1               6
1                 10                      3               3              10
2                  7                      3               3               0
3                  8                      3               3               8
4                  6                      3               3               2

  YearsInCurrentRole  YearsSinceLastPromotion  YearsWithCurrManager
0                  4                        0                     5
1                  7                        1                     7
2                  0                        0                     0
3                  7                        3                     0
4                  2                        2                     2

[5 rows x 35 columns]
      Age Attrition     BusinessTravel  DailyRate              Department  \
1465   36        No  Travel_Frequently        884  Research & Development
1466   39        No      Travel_Rarely        613  Research & Development
1467   27        No      Travel_Rarely        155  Research & Development
1468   49        No  Travel_Frequently       1023                   Sales
1469   34        No      Travel_Rarely        628  Research & Development
```

```
     DistanceFromHome  Education EducationField  EmployeeCount  \
1465               23          2        Medical              1
1466                6          1        Medical              1
1467                4          3  Life Sciences              1
1468                2          3        Medical              1
1469                8          3        Medical              1

     EmployeeNumber  ...  RelationshipSatisfaction StandardHours  \
1465           2061  ...                         3            80
1466           2062  ...                         1            80
1467           2064  ...                         2            80
1468           2065  ...                         4            80
1469           2068  ...                         1            80

     StockOptionLevel  TotalWorkingYears  TrainingTimesLastYear  \
1465                1                 17                      3
1466                1                  9                      5
1467                1                  6                      0
1468                0                 17                      3
1469                0                  6                      3

     WorkLifeBalance  YearsAtCompany  YearsInCurrentRole  \
1465               3               5                   2
1466               3               7                   7
1467               3               6                   2
1468               2               9                   6
1469               4               4                   3

     YearsSinceLastPromotion  YearsWithCurrManager
1465                       0                     3
1466                       1                     7
1467                       0                     3
1468                       0                     8
1469                       1                     2

[5 rows x 35 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       1470 non-null   int64
 1   Attrition                 1470 non-null   object
```

```
 2   BusinessTravel            1470 non-null   object
 3   DailyRate                 1470 non-null   int64
 4   Department                1470 non-null   object
 5   DistanceFromHome          1470 non-null   int64
 6   Education                 1470 non-null   int64
 7   EducationField            1470 non-null   object
 8   EmployeeCount             1470 non-null   int64
 9   EmployeeNumber            1470 non-null   int64
 10  EnvironmentSatisfaction   1470 non-null   int64
 11  Gender                    1470 non-null   object
 12  HourlyRate                1470 non-null   int64
 13  JobInvolvement            1470 non-null   int64
 14  JobLevel                  1470 non-null   int64
 15  JobRole                   1470 non-null   object
 16  JobSatisfaction           1470 non-null   int64
 17  MaritalStatus             1470 non-null   object
 18  MonthlyIncome             1470 non-null   int64
 19  MonthlyRate               1470 non-null   int64
 20  NumCompaniesWorked        1470 non-null   int64
 21  Over18                    1470 non-null   object
 22  OverTime                  1470 non-null   object
 23  PercentSalaryHike         1470 non-null   int64
 24  PerformanceRating         1470 non-null   int64
 25  RelationshipSatisfaction  1470 non-null   int64
 26  StandardHours             1470 non-null   int64
 27  StockOptionLevel          1470 non-null   int64
 28  TotalWorkingYears         1470 non-null   int64
 29  TrainingTimesLastYear     1470 non-null   int64
 30  WorkLifeBalance           1470 non-null   int64
 31  YearsAtCompany            1470 non-null   int64
 32  YearsInCurrentRole        1470 non-null   int64
 33  YearsSinceLastPromotion   1470 non-null   int64
 34  YearsWithCurrManager      1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
0       5993
1       5130
2       2090
3       2909
4       3468
       ...
1465    2571
1466    9991
1467    6142
```

1468   5390
1469   4404
Name: MonthlyIncome, Length: 1470, dtype: int64
<class 'pandas.core.series.Series'>
Mean: 6502.931292517007
Mode:
0   2342
Name: MonthlyIncome, dtype: int64
Variance: 22164857.07151842
Standard Deviation: 4707.956783097995

	Age	DailyRate	DistanceFromHome	Education	EmployeeCount
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.0
mean	36.923810	802.485714	9.192517	2.912925	1.0
std	9.135373	403.509100	8.106864	1.024165	0.0
min	18.000000	102.000000	1.000000	1.000000	1.0
25%	30.000000	465.000000	2.000000	2.000000	1.0
50%	36.000000	802.000000	7.000000	3.000000	1.0
75%	43.000000	1157.000000	14.000000	4.000000	1.0
max	60.000000	1499.000000	29.000000	5.000000	1.0

	EmployeeNumber	EnvironmentSatisfaction	HourlyRate	JobInvolvement
count	1470.000000	1470.000000	1470.000000	1470.000000
mean	1024.865306	2.721769	65.891156	2.729932
std	602.024335	1.093082	20.329428	0.711561
min	1.000000	1.000000	30.000000	1.000000
25%	491.250000	2.000000	48.000000	2.000000
50%	1020.500000	3.000000	66.000000	3.000000
75%	1555.750000	4.000000	83.750000	3.000000
max	2068.000000	4.000000	100.000000	4.000000

	JobLevel	...	RelationshipSatisfaction	StandardHours
count	1470.000000	...	1470.000000	1470.0
mean	2.063946	...	2.712245	80.0
std	1.106940	...	1.081209	0.0
min	1.000000	...	1.000000	80.0
25%	1.000000	...	2.000000	80.0
50%	2.000000	...	3.000000	80.0
75%	3.000000	...	4.000000	80.0
max	5.000000	...	4.000000	80.0

	StockOptionLevel	TotalWorkingYears	TrainingTimesLastYear
count	1470.000000	1470.000000	1470.000000
mean	0.793878	11.279592	2.799320
std	0.852077	7.780782	1.289271

min	0.000000	0.000000	0.000000
25%	0.000000	6.000000	2.000000
50%	1.000000	10.000000	3.000000
75%	1.000000	15.000000	3.000000
max	3.000000	40.000000	6.000000

	WorkLifeBalance	YearsAtCompany	YearsInCurrentRole	\
count	1470.000000	1470.000000	1470.000000	
mean	2.761224	7.008163	4.229252	
std	0.706476	6.126525	3.623137	
min	1.000000	0.000000	0.000000	
25%	2.000000	3.000000	2.000000	
50%	3.000000	5.000000	3.000000	
75%	3.000000	9.000000	7.000000	
max	4.000000	40.000000	18.000000	

	YearsSinceLastPromotion	YearsWithCurrManager
count	1470.000000	1470.000000
mean	2.187755	4.123129
std	3.222430	3.568136
min	0.000000	0.000000
25%	0.000000	2.000000
50%	1.000000	3.000000
75%	3.000000	7.000000
max	15.000000	17.000000

[8 rows x 26 columns]

	Age	Attrition	BusinessTravel	DailyRate	\
count	1470.000000	1470	1470	1470.000000	
unique	NaN	2	3	NaN	
top	NaN	No	Travel_Rarely	NaN	
freq	NaN	1233	1043	NaN	
mean	36.923810	NaN	NaN	802.485714	
std	9.135373	NaN	NaN	403.509100	
min	18.000000	NaN	NaN	102.000000	
25%	30.000000	NaN	NaN	465.000000	
50%	36.000000	NaN	NaN	802.000000	
75%	43.000000	NaN	NaN	1157.000000	
max	60.000000	NaN	NaN	1499.000000	

	Department	DistanceFromHome	Education	EducationField	\
count	1470	1470.000000	1470.000000	1470	
unique	3	NaN	NaN	6	
top	Research & Development	NaN	NaN	Life Sciences	

freq	961	NaN	NaN	606
mean	NaN	9.192517	2.912925	NaN
std	NaN	8.106864	1.024165	NaN
min	NaN	1.000000	1.000000	NaN
25%	NaN	2.000000	2.000000	NaN
50%	NaN	7.000000	3.000000	NaN
75%	NaN	14.000000	4.000000	NaN
max	NaN	29.000000	5.000000	NaN

	EmployeeCount	EmployeeNumber	...	RelationshipSatisfaction	\
count	1470.0	1470.000000	...	1470.000000	
unique	NaN	NaN	...	NaN	
top	NaN	NaN	...	NaN	
freq	NaN	NaN	...	NaN	
mean	1.0	1024.865306	...	2.712245	
std	0.0	602.024335	...	1.081209	
min	1.0	1.000000	...	1.000000	
25%	1.0	491.250000	...	2.000000	
50%	1.0	1020.500000	...	3.000000	
75%	1.0	1555.750000	...	4.000000	
max	1.0	2068.000000	...	4.000000	

	StandardHours	StockOptionLevel	TotalWorkingYears	\
count	1470.0	1470.000000	1470.000000	
unique	NaN	NaN	NaN	
top	NaN	NaN	NaN	
freq	NaN	NaN	NaN	
mean	80.0	0.793878	11.279592	
std	0.0	0.852077	7.780782	
min	80.0	0.000000	0.000000	
25%	80.0	0.000000	6.000000	
50%	80.0	1.000000	10.000000	
75%	80.0	1.000000	15.000000	
max	80.0	3.000000	40.000000	

	TrainingTimesLastYear	WorkLifeBalance	YearsAtCompany	\
count	1470.000000	1470.000000	1470.000000	
unique	NaN	NaN	NaN	
top	NaN	NaN	NaN	
freq	NaN	NaN	NaN	
mean	2.799320	2.761224	7.008163	
std	1.289271	0.706476	6.126525	
min	0.000000	1.000000	0.000000	
25%	2.000000	2.000000	3.000000	

```
50%              3.000000       3.000000       5.000000
75%              3.000000       3.000000       9.000000
max              6.000000       4.000000      40.000000

       YearsInCurrentRole  YearsSinceLastPromotion  YearsWithCurrManager
count         1470.000000              1470.000000          1470.000000
unique                NaN                      NaN                  NaN
top                   NaN                      NaN                  NaN
freq                  NaN                      NaN                  NaN
mean             4.229252                 2.187755             4.123129
std              3.623137                 3.222430             3.568136
min              0.000000                 0.000000             0.000000
25%              2.000000                 0.000000             2.000000
50%              3.000000                 1.000000             3.000000
75%              7.000000                 3.000000             7.000000
max             18.000000                15.000000            17.000000

[11 rows x 35 columns]
Index(['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',
       'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',
       'EmployeeNumber', 'EnvironmentSatisfaction', 'Gender', 'HourlyRate',
       'JobInvolvement', 'JobLevel', 'JobRole', 'JobSatisfaction',
       'MaritalStatus', 'MonthlyIncome', 'MonthlyRate', 'NumCompaniesWorked',
       'Over18', 'OverTime', 'PercentSalaryHike', 'PerformanceRating',
       'RelationshipSatisfaction', 'StandardHours', 'StockOptionLevel',
       'TotalWorkingYears', 'TrainingTimesLastYear', 'WorkLifeBalance',
       'YearsAtCompany', 'YearsInCurrentRole', 'YearsSinceLastPromotion',
       'YearsWithCurrManager'],
      dtype='object')
[[41 'Yes' 'Travel_Rarely' ... 4 0 5]
 [49 'No' 'Travel_Frequently' ... 7 1 7]
 [37 'Yes' 'Travel_Rarely' ... 0 0 0]
 ...
 [27 'No' 'Travel_Rarely' ... 2 0 3]
 [49 'No' 'Travel_Frequently' ... 6 0 8]
 [34 'No' 'Travel_Rarely' ... 3 1 2]]
     Age Attrition     BusinessTravel DailyRate              Department  \
0     41       Yes      Travel_Rarely      1102                   Sales
1     49        No  Travel_Frequently       279  Research & Development
2     37       Yes      Travel_Rarely      1373  Research & Development
3     33        No  Travel_Frequently      1392  Research & Development
4     27        No      Travel_Rarely       591  Research & Development
...   ..       ...                ...       ...                     ...
1465  36        No  Travel_Frequently       884  Research & Development
```

```
1466  39       No     Travel_Rarely      613  Research & Development
1467  27       No     Travel_Rarely      155  Research & Development
1468  49       No  Travel_Frequently     1023                 Sales
1469  34       No     Travel_Rarely      628  Research & Development

     DistanceFromHome  Education EducationField  EmployeeCount  EmployeeNumber  \
0                   1          2  Life Sciences              1               1
1                   8          1  Life Sciences              1               2
2                   2          2         Other               1               4
3                   3          4  Life Sciences              1               5
4                   2          1       Medical               1               7
...               ...        ...           ...            ...             ...
1465               23          2       Medical               1            2061
1466                6          1       Medical               1            2062
1467                4          3  Life Sciences              1            2064
1468                2          3       Medical               1            2065
1469                8          3       Medical               1            2068

      ... RelationshipSatisfaction StandardHours  StockOptionLevel  \
0     ...                       1            80                 0
1     ...                       4            80                 1
2     ...                       2            80                 0
3     ...                       3            80                 0
4     ...                       4            80                 1
...   ...                     ...           ...               ...
1465  ...                       3            80                 1
1466  ...                       1            80                 1
1467  ...                       2            80                 1
1468  ...                       4            80                 0
1469  ...                       1            80                 0

     TotalWorkingYears  TrainingTimesLastYear  WorkLifeBalance  YearsAtCompany  \
0                    8                      0                1               6
1                   10                      3                3              10
2                    7                      3                3               0
3                    8                      3                3               8
4                    6                      3                3               2
...                ...                    ...              ...             ...
1465                17                      3                3               5
1466                 9                      5                3               7
1467                 6                      0                3               6
1468                17                      3                2               9
1469                 6                      3                4               4
```

	YearsInCurrentRole	YearsSinceLastPromotion	YearsWithCurrManager
0	4	0	5
1	7	1	7
2	0	0	0
3	7	3	0
4	2	2	2
...	...	...	...
1465	2	0	3
1466	7	1	7
1467	2	0	3
1468	6	0	8
1469	3	1	2

[1470 rows x 35 columns]

RESULT:

The Pandas library was successfully used to perform data manipulation and analysis. DataFrames were used to read, organize, filter, and process data efficiently for data science applications.